

Raphaël Tinarrage

Curriculum vitae

Institute of Science and Technology Austria
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🌐 [raphaeltinarrage.github.io](https://github.com/raphaeltinarrage)



[ORCID](#) — [ResearcherID](#) — [ResearchGate](#) — [Google Scholar](#) — [arXiv](#) — [theses.fr](#) — [Lattes](#) — [GitHub](#) — [YouTube](#)

Academic positions

- 2024–present **Postdoc**, Institute of Science and Technology Austria (ISTA), Klosterneuburg
Classifying spaces in Topological Data Analysis, in Uli Wagner's group
- 2021–2024 **Postdoc**, Fundação Getulio Vargas – Escola de Matemática Aplicada (FGV EMAP), Rio de Janeiro
Theory and applications of Topological Data Analysis, supervised by César Camacho

Education

- 2017–2020 **PhD**, Inria Saclay and Laboratoire de Mathématiques d'Orsay
Topological inference from measures and vector bundles — [HAL:tel-02970491](#)
Supervised by Frédéric Chazal and Marc Glisse
[Manuscript](#) — [slides](#) — [reports](#)
- 2014–2017 **Master's degree**, Université Paris-Saclay
M2 research – Mathematics for life sciences
M1 – Fundamental and applied mathematics & Magistère de mathématiques 2nd year
- 2015–2016 **Master's degree**, École Normale Supérieure Paris-Saclay
M2 FESUP – Preparation for the *agrégation* examination (teaching diploma)
- 2013–2014 **Bachelor's degree**, Université Paris-Saclay
L3 – Fundamental and applied mathematics & Magistère de mathématiques 1st year
- 2011–2013 **Classes préparatoires**, Lycée Camille Pissarro, Pontoise
MPSI & MP

Examinations, competitions, fellowships

- 11/2024 **ISTA-Fellow: Postdoctoral Program**, Institute of Science and Technology Austria
Competitive [two-year fellowship](#) supporting research
- 11/2023 **Competition for Professor Adjunto**, Universidade do Estado do Rio de Janeiro (UERJ)
[Rank: 1st](#) (out of [10 participants](#))
- 01/2021 **Institutional Research Fellowship (bolsa de pesquisa)**, FGV EMAP, Brazil
Fellowship supporting research and teaching, held 42 months
- 07/2016 **Agrégation externe de mathématiques**, France
Competitive examination for teacher recruitment. [Rank: 68th](#) (out of 1896 participants)
- 09/2015 **Competitive entrance examination**, École normale supérieure Paris-Saclay
Admitted as one of [11 successful candidates](#) (second concours, série mathématiques).

Teaching

- 2024 **Vector Calculus**, FGV EMAP, Rio de Janeiro
2nd year undergraduate course (30 hours)
[Course webpage](#) — [notes](#) (original document, 180 pages, in Portuguese)
- 2023 **General and Combinatorial Topology**, FGV EMAP, Rio de Janeiro
Summer course for undergraduate and master's students (26 hours)
[Course webpage](#) — [notes](#) (original document, 95 pages, in English)
- 2021 **Topological Data Analysis with Persistent Homology**, FGV EMAP, Rio de Janeiro
Summer course for undergraduate and master's students (22 hours)
[Course webpage](#) — [notes](#) (original document, 97 pages, in English) — [videos](#)
- 2017–2020 **Statistical Interpretation of Data**, UE M331, L3 MINT, Université Paris-Saclay, Orsay
Teaching fellow for undergraduate students
- 2017–2020 **Modeling Project**, UE M326, L3 MINT, Université Paris-Saclay, Orsay
Teaching fellow for undergraduate students
- 2017–2019 **Ordinary Differential Equations**, UE M257, L2 BC, Université Paris-Saclay, Orsay
Teaching fellow for undergraduate students — [Course webpage](#)

Mentoring

- 2026 **Supervision of a BSc summer internship**, ISTA, Klosterneuburg
Topic: Integral persistent homology of torus knots
Student: Damien Bellon
- 2025–2026 **Co-supervision of a bachelor's thesis**, FGV EMAp, Rio de Janeiro
Topic: Specialization of deep layers in language models
Student: Zuilho Segundo
- 2022–2023 **Supervision of a team of bachelor's students**, FGV EMAp, Rio de Janeiro
Topic: Fine-tuning legal language models via annotations
Students: Livia Cales, Victoria Cury, Samanta Duarte, Clara Lopes, Eduardo Portol, João Meirelles, Ana Rosenberg, and Helena Torres
- 2021–2023 **Supervision of a master's thesis**, FGV EMAp, Rio de Janeiro
Topic: Data analysis of symmetries with Lie groups
Student: Henrique Ennes
- 2021–2023 **Supervision of a team of bachelor's students**, FGV EMAp, Rio de Janeiro
Topic: Machine learning for *Súmulas Vinculantes*
Students: Beatriz Chagas, Carla Damian, Ana Clara Jaccoud, and Pedro de Oliveira
- 2017–2020 **MATh.en.JEANS Workshop**, Collège Alain Fournier, Orsay
Mathematics outreach in college — [webpage](#)

Journal articles

- 05/2026 **Train-Free Segmentation in MRI with Cubical Persistent Homology**
Anton François, **Raphaël Tinarrage**
Journal of Mathematical Imaging and Vision **68**, 20 (2026)
[DOI:10.1007/s10851-026-01300-1](#) — [arXiv:2401.01160](#)
22 double column pages, in English
- 09/2025 **LieDetect: Detection of representation orbits of compact Lie groups from point clouds**
Henrique Ennes, **Raphaël Tinarrage**
Foundations of Computational Mathematics (2025)
[DOI:10.1007/s10208-025-09728-4](#) — [arXiv:2309.03086](#)
136 pages, in English
- 05/2025 **Empirical analysis of binding precedent efficiency in Brazilian Supreme Court via case classification**
Raphaël Tinarrage, Henrique Ennes, Lucas Resck, Lucas T. Gomes, Jean R. Ponciano, Jorge Poco
Artificial Intelligence and Law (2025)
[DOI:10.1007/s10506-025-09458-6](#) — [arXiv:2407.07004](#)
67 pages, in English
- 01/2025 **ZigzagNetVis: Suggesting temporal resolutions for graph visualization using zigzag persistence**
Raphaël Tinarrage, Jean Ponciano, Cláudio Linhares, Agma Traina, Jorge Poco
IEEE Transactions on Visualization and Computer Graphics **31**(10), 6852–6869 (2025)
[DOI:10.1109/TVCG.2025.3528197](#) — [arXiv:2304.03828](#)
18+7 (supplementary) double column pages, in English
- 02/2023 **Recovering the homology of immersed manifolds**
Raphaël Tinarrage
Discrete & Computational Geometry **69**, 659–744 (2023)
[DOI:10.1007/s00454-022-00409-5](#) — [arXiv:1912.03033](#)
86 pages, in English
- 11/2021 **Computing persistent Stiefel-Whitney classes of line bundles**
Raphaël Tinarrage
Journal of Applied and Computational Topology **6**, 65–125 (2022)
[DOI:10.1007/s41468-021-00080-4](#) — [arXiv:2005.12543](#)
61 pages, in English

Conference articles

- 06/2026 **Simplicial approximation to CW complexes with spherical Delaunay triangulations**
Raphaël Tinarrage
International Symposium on Computational Geometry (SoCG 2026)
[DOI:10.4230/LIPIcs.SoCG.2026.93](#) — [arXiv:2112.07573](#)
53 pages, in English

- 09/2022 **O impacto da Súmula Vinculante 26 na diminuição de demanda similar no STF: uma análise quantitativa por modelos de ML**
Beatriz Sabdin Chagas, Carla Marcondes Damian, **Raphaël Tinarrage**
XI Encontro Internacional do CONPEDI Chile, 82–103 (2022)
[ISBN:978-65-5648-559-1](#)
22 pages, in Portuguese
- 09/2022 **Progressão de regime em crimes hediondos no Supremo Tribunal Federal: uma análise empírica pela Súmula Vinculante 26**
Ana Clara Macedo Jaccoud, Pedro Burlini de Oliveira, **Raphaël Tinarrage**
XI Encontro Internacional do CONPEDI Chile, 399–418 (2022)
[ISBN:978-65-5648-569-0](#)
20 pages, in Portuguese
- 06/2020 **DTM-based filtrations**
Hiroyasu Anai, Frédéric Chazal, Marc Glisse, Yuichi Ike, Hiroyasu Inakoshi, **Raphaël Tinarrage**, Yuhei Umeda
International Symposium on Computational Geometry (SoCG 2019) — [DOI:10.4230/LIPIcs.SoCG.2019.58](#)
Abel Symposia 15, Springer, Cham. (2020) — [DOI:10.1007/978-3-030-43408-3_2](#)
[arXiv:1811.04757](#)
33 pages, in English

Preprints

- 06/2026 **Sharp approximate Carathéodory theorem and application to iterated Delaunay refinement**
Raphaël Tinarrage
[arXiv:2606.25854](#)
61 pages, in English

Posters

- 07/2026 **Simplicial approximation with Delaunay complexes**, [FoCM 2026](#), Universität Wien
[Poster](#)
- 06/2022 **Simplicial approximation to CW-complexes in practice**, [ATMCS 2022](#), University of Oxford
[Poster](#)
- 06/2018 **DTM-filtrations**, [ATMCS 2018](#), IST Austria
[Poster](#)

Selected invited talks

- 11/2025 **London-Oxford-Paris TDA Seminar**, City St George's
Simplicial approximation, Delaunay triangulations, and the list homomorphism problem
[Slides](#) — [program](#)
- 06/2025 **Applied Algebraic Topology Research Network**, Online
Detection of representation orbits of compact Lie groups from point clouds
[Slides](#) — [video](#)
- 11/2022 **Seminário ICMC**, Universidade de São Paulo (USP), São Carlos
TDA para escolha de resolução temporal na visualização de grafos
[Slides](#)
- 12/2020 **Applied Algebraic Topology Research Network**, Online
Persistent Stiefel-Whitney classes
[Slides](#) — [video](#)

Selected contributed talks

- 06/2026 **International Symposium on Computational Geometry 2026**, Rutgers University
Simplicial approximation to CW complexes with spherical Delaunay triangulations
[Slides](#) — [program](#)
- 02/2025 **Infinite-dimensional Geometry Conference**, Erwin Schrödinger Institute (ESI)
Train-Free Segmentation in MRI with Cubical PH
[Slides](#)
- 07/2024 **XXIII Encontro Brasileiro De Topologia**, Universidade Federal da Bahia (UFBA)
Classifying spaces in TDA
[Slides](#) — [program](#)
- 01/2023 **Summer School on Data Science**, FGV EMap
I. From Topology to Data Analysis: [Slides](#) — [video](#)
II. Homological inference: [Slides](#) — [video](#)
III. Persistent Homology: [Slides](#) — [video](#)